

Education

BSc (Hons) Computer Science **2022-2025** **University Of Manchester**
Grade: 1st Class (4.0 GPA) **Dissertation:** *Category-theoretic framing of register machine semantics* (ranked 5th/200)
Courses: *Machine Learning, Compilers, GPU & Parallel Programming, Advanced Algorithms & Data Structures*

Experience

C++ Software Engineer **June 2025-**
TuringDB **London**

- Leading engineer on the world's fastest analytical graph database engine: vectorised and column-oriented; in C++23
- Reduced CPU cycle counts by 66%** on hot-path DB operations; motivated by **perf** and **callgrind** profiling
- Researched, designed, and implemented novel **cache-conscious** join algorithms; **50x+ faster** than competitors
- Tailored implementations of **lock free** and **persistent** datastructures; delivered stable 99th percentile latency
- Developing a custom **MLIR dialect** for relational algebras, storage layer access, and query optimisation passes
- Optimised memory access patterns of all stages of **query interpretation**: parsing; semantic analysis; execution
- Leveraged **data-orientated design** in implementation of **entity component systems** at the storage layer
- Lead **Nix** cross-platform, cross-architecture packaging and development/build environments for reproducibility

Freelance Python Software Engineer **2023-2025**
Optimistics **London**

Projects

Mini Operating System for ARM microcontroller <https://github.com/cyrusknopf/arm-assembly-os>

- Engineered core OS functions, extending them to develop a program with multimodal I/O interaction with the user
- Added support for software debouncing with custom, hand-written **interrupt handlers** and **supervisor calls**
- Synthesised **Verilog** onto an **FPGA** to uniquely extend functionality and facilitate more complex programs

PRISM - Portfolio Risk, Investment & Strategy Management <https://github.com/cyrusknopf/PRISM-hackathon>

- Orchestrated a university hackathon with **100+ participants** by designing a trading platform using **Go** and **Python**
- Engineered a challenge requiring **NLP** analysis and portfolio selection using an optimised, bespoke, self-hosted **LLM**
- Built a **containerised infrastructure** (Docker, Go, Python, Next.js, Postgres) handling **80,000+** requests per second
- Implemented a multi-metric evaluation system as a function of Sharpe, Sortino, and diversity for portfolio evaluation

Monte Carlo Tree Search Chess Agent <https://github.com/cyrusknopf/yggdrasil>

- Built a bitboard agent with my own implementation of the Monte Carlo tree search algorithm in modern **C++23**
- Data-driven optimisations motivated by **microbenchmarking** with Google Benchmark resulting in **10x** speedups
- Identified hot-path and key functions to target optimisations using **profiling** tools such as **callgrind**

Optiver Hackathon Trading Bot <https://github.com/cyrusknopf/optibook-challenge/>

- Developed an algorithmic trading bot in Python for a mock financial exchange, accumulating **\$10,000/hr** at peak
- Leveraged arbitrage across **2 markets**, managing a **delta neutral** portfolio over 3 products and 2 baskets

Functional Data Structures <https://github.com/cyrusknopf/dust>

- Implemented common (e.g. doubly linked list) and esoteric (e.g. zipper) immutable **data structures** in **OCaml**
- Evaluation and validation of implementation with microbenchmarking (**Bench**) and expect-tests (**ppx_expect**)

Skills

C++, Nix, Linux CLI, Bash, Git, Linux perf, Valgrind, GDB, Intel VTune, ARM Assembly, Vim

Relevant Interests

Low latency C++, programming language design, lambda calculus, compilers, functional programming